

Lecture 5: Dynamic Taxation I: Capital Taxation

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Taxation with Idiosyncratic shocks (Aiyagari, JPE 1995)

- ▶ infinitely lived household
- ▶ Each household receives an idiosyncratic shock θ_t that follows some non-degenerate Markov process.
- ▶ Let $\theta^t = (\theta_1, \dots, \theta_t)$ is a history of shocks that each household received up to period t
- ▶ Households solve

$$v(b_0, \theta_0; \{\tau_t^k, \tau_t^l\}) = \max_{c, k, b, l} \mathbb{E}_0 \sum \beta^t u(c_t, l_t)$$

s.t.

$$c(\theta^t) + k_{t+1}(\theta^t) + b_{t+1}(\theta^t) \leq (1 - \tau_t^l) w_t \theta_t l_t(\theta^t) + (1 - \tau_t^k)(1 + r_t - \delta)(k_t(\theta^{t-1}) + b_t(\theta^{t-1}))$$

$$k_{t+1}(\theta^t) + b_{t+1}(\theta^t) \geq D_{t+1}(\{\tau_t^k, \tau_t^l\})$$

- ▶ Here $D_{t+1}(\{\tau_t^k, \tau_t^l\})$ is the smallest present value of income that a household can make from period $t + 1$ onwards.

Firms and Government

► Firms

$$\max F(K_t, L_t) - r_t K_t - w_t L_t$$

► Government picks $\{\tau_t^k, \tau_t^l, B_t, g_t\}$ to maximize

$$\mathbb{E}_0 \sum \beta^t u(c_t, l_t) + U(g_t)$$

with budget constraint

$$g_t + B_t = \tau_t^k(1 + r_t - \delta)(K_t + B_t) + \tau_t^l w_t L_t$$

► Feasibility

$$\int b_t(\theta^t) dF(\theta^t) + B_t = 0$$

$$\int \theta_t l_t(\theta^t) dF(\theta^t) = L_t$$

$$\int k_t(\theta^t) dF(\theta^t) = K_t$$

$$\int c_t(\theta^t) dF(\theta^t) + g_t + K_{t+1} = F(K_t, L_t) + (1 - \delta)K_t$$

► Redefine variables $\bar{w}_t \equiv (1 - \tau_t^l)w_t$, $\bar{r}_t \equiv (1 - \tau_t^k)(1 + r_t - \delta)$, $a_t \equiv k_t + b_t$

► Then the Ramsey planner can maximize over

$$\{w_t, r_t, \bar{w}_t, \bar{r}_t, K_t, B_t, L_t, a_t(\theta^t), c_t(\theta^t), l_t(\theta^t)\} \text{ instead of } \\ \{w_t, r_t, \tau_t^l, \tau_t^k, K_t, B_t, L_t, b_t(\theta^t), k_t(\theta^t), c_t(\theta^t), l_t(\theta^t)\}$$

► Households problem

$$v(a_0, \theta_0; \{\tau_t^k, \tau_t^l\}) = \max_{c, a, l} \mathbb{E}_0 \sum \beta^t u(c_t, l_t)$$

s.t.

$$c(\theta^t) + a_{t+1}(\theta^t) \leq \bar{w}_t \theta_t l_t(\theta^t) + \bar{r}_t a_t(\theta^{t-1}) \\ a_{t+1}(\theta^t) \geq D_{t+1}(\{\bar{r}_t, \bar{w}_t\})$$

► We define optimal choices recursively as

$$c_t = c_t(a_t, \theta_t; \{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})$$

$$l_t = l_t(a_t, \theta_t; \{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})$$

$$a_{t+1} = a_{t+1}(a_t, \theta_t; \{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})$$

Household Problem

- This allows us to define functions

$$L_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty}), C_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty}), B_t = A_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty}) + K_t$$

- Important: if we have $\{a_t(\theta^t)\}_{\theta^t}, K_t$, then any $\{k_t(\theta^t), b_t(\theta^t)\}_{\theta^t}$ distribution that satisfies $k_t(\theta^t) + b_t(\theta^t) = a_t(\theta^t)$ and $\int k_t(\theta^t) = K_t$ is an equilibrium which gives the same welfare
- Therefore we can independently pick $\{a_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})\}$ and K_t , and can back out $\{b_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})\}, \{k_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})\}$ and government debt $\{B_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})\}$ as residuals.

Step back

- ▶ So far, not much progress
 - ▶ we have no idea how functions $\{L_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})\}$, $\{C_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})\}$, $\{A_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})\}$ depend on $\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty}$
- ▶ We could proceed as before and take all the FOCs but it is very hard to get any insight
 - ▶ we have infinitely many histories θ^t and corresponding constraints
 - ▶ distribution of asset holdings $\{a_t\}$ and as a result policy rules $c_t(a_t, \theta_t; \{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})$, etc are very hard to characterize
- ▶ This is a very general problem with incomplete market models
 - ▶ distribution of asset holdings and labor supply depend on current and future prices in a complicated way
 - ▶ prices are endogenous, function of the distribution of asset holdings and labor supply
 - ▶ even without taxation, equilibrium is a complicated fixed point problem which is hard to solve even numerically (see Krusell and Smith (JPE, 1998))
 - ▶ here, in addition we impose an outer layer of choosing optimal taxes

Partial characterization

- ▶ Let's consider a partial characterization. Suppose the planner took $l_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty}), c_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})$ as given
- ▶ Ramsey problem is

$$\max_{\bar{r}, \bar{w}, K, g} \mathbb{E}_0 \sum \beta^t u(c_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty}), l_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})) + U(g_t)$$

s.t

$$C_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty}) + g_t + K_{t+1} = F(K_t, L_t(\{\bar{r}_t, \bar{w}_t\}_{t=0}^{\infty})) + (1 - \delta)K_t$$

- ▶ Let's characterize capital taxation in the steady state

First order conditions

- ▶ $[g_t] : \beta^t U'(g_t) = \lambda_t$
- ▶ $[K_{t+1}] : \lambda_t = (1 + F_k(t+1) - \delta)$
- ▶ In steady state: prices and aggregate quantities are constant, individual quantities are not.
- ▶ In steady state aggregate capital stock is "undistorted"

$$(1 + F_k(ss) - \delta) = \frac{1}{\beta}$$

Implication for taxes

- Remember consumer's problem

$$v(a_0, \theta_0; \{\tau_t^k, \tau_t^l\}) = \max_{c, a, l} \mathbb{E}_0 \sum \beta^t u(c_t, l_t)$$

s.t.

$$\begin{aligned} c(\theta^t) + a_{t+1}(\theta^t) &\leq \bar{w}_t \theta_t l_t(\theta^t) + \bar{r}_t a_t(\theta^{t-1}) \\ a_{t+1}(\theta^t) &\geq D_{t+1}(\{\bar{r}_t, \bar{w}_t\}) \end{aligned}$$

- FOCs w.r.t a_{t+1} imply Euler equation

$$u_c(t) \geq \bar{r} \beta \mathbb{E}_t u_c(t+1)$$

Martingale Convergence Theorem

- ▶ Martingale convergence theorem: one of the most important mathematical results for characterization of long-run distributions
- ▶ Stochastic process M_t is a
 - ▶ martingale if $M_t = \mathbb{E}_t M_{t+1}$ for all t
 - ▶ submartingale if $M_t \leq \mathbb{E}_t M_{t+1}$ for all t
 - ▶ supermartingale if $M_t \geq \mathbb{E}_t M_{t+1}$ for all t

Theorem

Suppose M_t is a martingale (or submartingale, or supermartingale) such that $\sup_t \mathbb{E}_0 |M_t| < \infty$ (the expectations are bounded). Then M_t converges almost surely to a finite limit.

Application of MCT

- ▶ Suppose $\bar{r} \geq 1/\beta$. Then

$$u_c(t) \geq \mathbb{E}_t u_c(t+1)$$

- ▶ Therefore, $u_c(T)$ is a supermartingale.

- ▶ $u_c(t)$ is a bounded supermartingale since $u_c(t) \geq 0$ implies

$$u_c(0) \geq \mathbb{E}_t u_c(t+1) \geq 0$$

- ▶ Therefore, if $\bar{r} \geq 1/\beta$ then $u_c(t)$ converges to a finite limit.

Implications I

- Suppose that $u_c(t) \rightarrow u_c^* > 0$. Then $c_t \rightarrow c^* < \infty$. This c^* should satisfy a budget constraint. For $\varepsilon > 0, \exists T, \forall t \geq T$

$$c^* + a_{t+1}(\theta^t) \leq \bar{w}_t \theta_t l_t(\theta^t) + \bar{r}_t a_t(\theta^{t-1})$$

- Different realizations of θ_t will lead to different l_t and different $\bar{w}_t \theta_t l_t(\theta^t)$
- Consider a long sequence of high and low realizations of $\bar{w}_t \theta_t l_t(\theta^t)$ so a_t should diverge to $\pm\infty$ which would violate the natural debt limit or transversality constraint. (or TVC)

Implications II

- ▶ Alternatively, suppose that $u_c(t) \rightarrow 0$ so that c_t diverges to ∞
- ▶ Then it must be true that a_t diverges to infinity

- ▶ Since

$$K_t + B_t = \int a_t$$

it must be true that $K_t + B_t$ diverges to infinity

- ▶ but K_t is bounded because there is a maximum sustainable capital stock in the economy
- ▶ B_t is bounded by present value of tax revenues on capital and labor
- ▶ Which leads to a contradiction.

Implications III

- ▶ This arguments show that $u_c(t)$ cannot be a martingale, and therefore

$$\bar{r} < 1/\beta$$

- ▶ This implies at a steady state

$$\begin{aligned} 1 + F_k(K_{ss}, L_{ss}) - \delta &= 1/\beta \\ &> \bar{r} = (1 - \tau_{ss}^k)(1 + r_{ss} - \delta) \\ &= (1 - \tau_{ss}^k)(1 + f_K(K_{ss}, L_{ss}) - \delta) \end{aligned}$$

- ▶ Therefore the optimal tax on capital is positive

$$\tau_{ss}^k > 0$$

Discussion

- ▶ In the presence of uninsurable idiosyncratic shocks, consumer has a precautionary motive to save, in addition to the regular consumption smoothing forces
 - ▶ for any finite amount of assets there is always a positive probability of hitting borrowing limit eventually
 - ▶ if $\bar{r} \geq 1/\beta$ consumer faces no (first order) losses by slightly increasing his savings and moving away from borrowing limit
 - ▶ finite distribution of assets is possible only with $\bar{r} < 1/\beta$
- ▶ Ramsey planner wants to be able to transfer resource between the periods with the shadow cost being equal to rate of discounting, which lead to $1 + F_k - \delta = 1/\beta$

The original result

- ▶ W. Rogerson (1985), “Repeated Moral Hazard,” *Econometrica* 53 (1985), 69–76.
- ▶ Moral hazard model.
- ▶ Two periods $t = 0, 1$
 - ▶ effort in first period e_0
 - ▶ stochastic output in second period $y_1 = \theta_1$ with density $f(\theta_1|e_0)$
 - ▶ consumption in both periods c_0 and $c_1(\theta_1)$
 - ▶ linear savings technology with rate of return R

- ▶ Separable utility

$$u(c_0) - h(e_0) + \beta \int u(c_1(\theta_1))f(\theta_1|e_0)$$

- ▶ Incentive compatibility of $\{c_0, e_0, c_1(\theta_1)\}$ requires

$$u(c_0) - h(e_0) + \beta \int u(c_1(\theta_1))f(\theta_1|e_0) \geq u(c_0) - h(e'_0) + \beta \int u(c_1(\theta_1))f(\theta_1|e'_0)$$

Planning Problem

► Planning Problem

$$\min C(u_0) + \frac{1}{R} \int [C(u_1(\theta_1)) - \theta_1] f(\theta_1 | e_0)$$

s.t. incentive compatibility and promise-keeping

$$u_0 - h(e_0) + \beta \int u(c_1(\theta_1)) f(\theta_1 | e_0) \geq u_0 - h(e'_0) + \beta \int u(c_1(\theta_1)) f(\theta_1 | e'_0)$$

$$u_0 - h(e_0) + \beta \int u(c_1(\theta_1)) f(\theta_1 | e_0) \geq \underline{U}$$

► If agents could save at R then we would have the Euler equation

$$u'(c_0) = \beta R \int u'(c_1(\theta_1)) f(\theta_1 | e_0)$$

► We will show that this equation does not hold at the solution of the planning problem: there are savings distortions.

Proof

- ▶ Fix e_0 and consider variations in consumption/utility:

$$\hat{u}_0 = u_0 - \beta \Delta$$

$$\hat{u}_1(\theta_1) = u_1(\theta_1) + \Delta$$

- ▶ Preserves utility and incentive compatibility since for all e'_0 we have

$$\hat{u}_0 - h(e_0) + \beta \int u(c_1(\theta_1))f(\theta_1|e_0) = u_0 - h(e_0) + \beta \int u(c_1(\theta_1))f(\theta_1|e_0)$$

- ▶ The optimal allocation must be immune to such variations

$$0 = \arg \min_{\Delta} C(u_0 - \beta \Delta) + \frac{1}{R} \int [C(u_1(\theta_1) + \Delta) - \theta_1] f(\theta_1|e_0)$$

- ▶ Note a similarity with a savings problem, where Δ looks like an asset, $-C(-x)$ looks like a utility function, β^{-1} is the interest rate and R is the discount factor.

Proof

- ▶ The associated (FOC evaluated at $\Delta = 0$ is

$$C'(u_0) = \frac{1}{\beta R} \int C'(u_1(\theta_1)) f(\theta_1 | e_0)$$

i.e

$$\frac{1}{u'(c_0)} = \frac{1}{\beta R} \int \frac{1}{u'(c_1(\theta_1))} f(\theta_1 | e_0)$$

This is the Inverse Euler equation.

- ▶ As long as the variance of $c_1(\theta_1)$ is strictly positive, then concavity of marginal utility and Jensen's inequality implies that the Euler equation is violated

$$u'(c_0) < \beta R \int u'(c_1(\theta_1)) f(\theta_1 | e_0)$$

- ▶ Agents are “savings-constrained”. We have

$$u'(c_0) = \beta R(1 - \tau^s) \int u'(c_1(\theta_1)) f(\theta_1 | e_0)$$

with a positive savings wedge (implicit tax on savings/capital) $\tau^s > 0$

Application to Dynamic Mirrlees Model

- ▶ M. Golosov, N. Kocherlakota, and A. Tsyvinski, "Optimal Indirect and Capital Taxation," Review of Economic Studies 70 (2003), 569-587.
- ▶ Two periods $t = 0, 1$
 - ▶ work in second period $y_1(\theta_1)$
 - ▶ consumption in both periods c_0 and $c_1(\theta_1)$
 - ▶ linear savings technology with rate of return R
 - ▶ Separable utility

$$u(c_0) + \beta \int [u(c_1(\theta_1)) - h(y_1(\theta_1), \theta_1)] f(\theta_1 | e_0)$$

- ▶ Similar planning problem...same optimality condition, Inverse Euler equation

$$\frac{1}{u'(c_0)} = \frac{1}{\beta R} \int \frac{1}{u'(c_1(\theta_1))} f(\theta_1 | e_0)$$

Same implications for the savings wedge.

Discussion

- ▶ The uncertainty in types over time and the inability to control labor supply force the planner to impose a distortion on savings to improve the provision of incentives to work.
- ▶ Another way to see this is that the desired labor supply at the optimal allocation is incompatible with free savings.
- ▶ Increasing savings in period t increases disposable income in period $t + 1$.
- ▶ Unless utility is quasilinear, this implies an income effect on labor supply, and the agent is then tempted to work less.
- ▶ More savings in period t and lower labor supply in period $t + 1$ are complements.
- ▶ Ruling out such a deviation requires discouraging savings below the level that would occur at the free-market rate.
- ▶ Quantitatively savings wedge small, and most of the insurance done with labor wedge... that will be our next class.

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